Midterm Written Report

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Research Question

- **Question:** Do 311 service requests in San Francisco's lower-income census tracts take longer to close than requests in higher-income tracts? Does that gap persist when we compare requests of the same type, submitted through the same channel? And does it follow tract racial and ethnic composition as well as income?
- **Area and unit:** City and County of San Francisco, census tracts.
- **Time frame:** from 2020 to 2025
- **Changes since the proposal:**
    a. Counted requests by type and by income group, because request volume affects backlog.
    b. The extra ACS tables (race and ethnicity, renters, limited English, poverty) are not added yet.
    c. 899 points that fall in no tract are dropped and logged, not moved to the nearest tract.

Data

- **Data source**: DataSF 311 Cases (vw6y-z8j6). requests opened in 2020 through 2025, downloaded month by month where the number of rows matched the API total.
- **311 requests**: 4,374,679 raw rows → 2,434,928 kept (55.7%). The rules are applied in this order. each row is counted as violating the first rule it violates.
- **Rows removed**:
    a. Test records: 546 (0.01%). Not a real request.
    b. Other types of request: 1,269,348 (29.0%). We analyze only 10 field-visit types.
    c. Closed as duplicate: 280,297 (6.4%). The closing time is not a service time.
    d. Transferred: 85,131 (1.9%). Action occurs outside of this request.

e. Administrative closure: 118,334 (2.7%). Closed due to process issues.
f. Not actionable: 58,777 (1.3%). Closed without field visit (invalid, canceled or missing information).
g. Impossible dates: 1,682 (0.04%). Closed prior to opening or without date.
h. No coordinates: 125,627 (2.9%). Cannot be mapped to a tract.
i. Outside of San Francisco: 9 (<0.01%), outside of the city's bounding box.

- **Fixes**: text trimming, dates interpreted in San Francisco time, (0,0) coordinates made NA; the category rename in June 2024 is mapped in category_map.csv; closure statuses coded according to status_rules.csv (exceptions include "not a duplicate").
- **Waiting time**: number of days between open and close. 73,033 requests (3.0% of kept) remained open at download and were therefore kept and flagged, along with their days since open.
- **Limitations**:
    a. Phone requests: high proportion for 2022–2024 without coordinates (51.5%, 74.2% and 29.4% vs. at most 1.1% in other years), hence underrepresented; we will compare results with and without them.
    b. Streetlights: no coordinates in any 2023 request, no streetlights in that year.
    c. Sewer: closing time median goes down to a few hours (2022 and on), likely a change in recording; was 176 days (2020) and 55 days (2021).
    d. Sidewalk and Curb: about 47% of closures have a date but no time, hence waiting time is known only to the day.
    e. Street and Sidewalk Cleaning: 70% of kept requests, comparing within each request types by income group.
- **ACS 2020–2024 5-year estimates (income B19013, population B01003), 244 tracts**:
    a. 10 missing-income codes set to missing, not zero.
    b. 10 tracts top-coded at $250,001, flagged.
    c. 29 tracts with CV above 0.30 (CV = MOE ÷ 1.645 ÷ estimate), considered unreliable.
    d. 3 tracts with zero population.

e. 205 tracts have a reliable income.

Methods

- **Download:**
    a. 311 requests from the DataSF API;
    b. ACS table and 2024 tract boundaries from the IPUMS NHGIS API.
- **Clean:** Data cleaning methods described above
- **Spatial join:** After performing data cleaning, converted each request's longitude and latitude into a point and projected both the points and tract boundaries to points and tracts We then used GeoPandas' spatial join with the within predicate to identify the census tract containing each request projected to **EPSG:2227** (NAD83 / California zone 3, feet); gpd.sjoin(predicate="within"). The join assigned 2,434,029 requests (99.96%) to San Francisco census tracts. The remaining 899 requests (0.04%) fell outside the available tract polygons. We preserved these records separately and excluded them from tract-level summaries. Unlike the nearest-tract assignment proposed initially, this approach avoids assigning uncertain locations to potentially incorrect tracts.

- **Chart 1: cleaned 311 requests by type**
    1. Used all 2,434,928 cleaned requests from 2020 to 2025.
    2. Counted how many requests from each request category.
    3. Divide each count by the total to get each type's share.
    4. Sorted the types from most to least

- **Chart 2: Response time by tract income**
    1. Used each tract's median household income from the ACS 2020–2024 5-year estimates.
    2. Kept the 203 tracts with a reliable income estimate
    3. Ranked these tracts by income and split them into four equal groups (quartiles)

4. Gave each request the income group of the tract it falls in, using the census tract ID (GEOID) from the spatial join.
5. For each request, response time is the number of days from when it was opened to when it was closed.
6. We took the median response time in each income group. We used the median because a few requests stay open for months and pull the average up.

- **Chart 3: Total requests per year by tract income**
    1. Same four income groups as in Chart 2
    2. Each requests the income group of its tract.
    3. Counted the requests in each income group for each year from 2020 to 2025.
    4. Shown in total counts
    5. One line per income group (with the year on the x-axis and the number of requests (in thousands) on the y-axis)

Early Results

- **Primary conclusion:** median times are not longer for low-income tracts: 18 hours in the lowest-income tracts compared to 21 hours in the highest.

- **Map:** Requests are concentrated on the eastern side (Downtown, Tenderloin, SoMa, Mission); the most active tract is found in the Mission and has 46,041 requests.

- **Chart 1:**
    a. Street and Sidewalk Cleaning is
        - 69.6% of requests
        - Graffiti 11.6%
        - Encampments 6.8%
        - each other type is 3%

- **Chart 2:**
    a. median days to close are
        - 0.76 (Q1)
        - 0.73 (Q2)
        - 0.76 (Q3)
        - 0.88 (Q4)
        - Means are 7.2 to 11.1 days
        - Requests still open: 2.5% in Q1 vs 4.1% in Q4.

- **Chart 3:** Q4 files the fewest requests every year. All groups dip in 2022 and rise from 2023. Q1 grew from 72,692 (2020) to 108,983 (2025). Q2 has the most requests because it has the most residents.

- **Surprise:** Yes, that is against our initial hypothesis. One reason may be the request mix: low-income tracts make more street cleaning requests, which are closed in less than a half day. With regard to request type, there are both faster and slower requests: Tree Maintenance is 14.9 days in Q1 and 8.8 in Q4.

Plan to Finish

- **Remaining work:**
    a. Add ACS race and ethnicity, renters, limited English and poverty.
    b. response time by income within each request type and channel
    c. Map of median response time by tract.
    d. Regression and race and ethnicity comparison.
    e. Website
    f. final write-up

- **Risks:**

a. Request mix: street cleaning dominates, so we will compare within each type.
b. Open requests: more are open in Q4, so we will use a survival model or report the share still open.
c. Missing phone locations in 2022–2023: we will report those years separately if results change.
d. Closed does not always mean fixed: we will avoid causal language.

- **Questions for feedback:**
  a. Is a within-type comparison enough to explain why low-income tracts close faster, or should we also control for location?
  b. Is a survival model needed for open requests, or is reporting the share still open enough?